32 Checkout, System Testing, and Startup

By Michael Cable

Topic Highlights

Instrumentation Commissioning
Software Testing
System Level Testing
Factory Acceptance Testing (FAT)
Site Acceptance Testing (SAT)
Safety Considerations

32.1 Introduction

Many automation professionals are involved in planning, specifying, designing, programming, and integrating instrumentation and controls required for process automation. In this section we will describe the various methods of testing the installation, integration, and operation at the component and system level. The goal in the end is to ensure the integrated system functions the way it was intended when everyone got together in the beginning and specified what they wanted the systems to do.

In an ideal world, we could wait until the entire system is ready for operation and perform all testing at the end. This would allow a much more efficient means of testing, with everything connected, communicating, and operational. However, we all know several problems would be uncovered that would lead to long startup delays. Uncovering the majority of these problems at the earliest opportunity eliminates many of these delays and can be corrected at a much lower cost.

An efficient means of testing the system can be developed, resulting in limited duplication of effort by properly planning, communication, and using a standardized approach to instrumentation and control system commissioning. Instrumentation and control system commissioning can be defined as a planned process by which instrumentation and control loops are methodically placed into service. Instrument commissioning can be thought of as building a case to prove the instrumentation and controls will perform as specified.

This chapter does not cover testing performed during software development, as this should be covered in the developer's software quality assurance procedures. However, a formal testing of the function blocks or program code, which will be described later in this section, should be performed and documented. This chapter does not cover documentation and testing required for equipment (for example, pumps, tanks, heat exchangers, filters, and air handling units). The scope for this section begins at the point when instruments are received, panels are installed, point-to-point wiring is completed, and systems have been turned over from construction.

The plan for testing described in this section must consider where the system is being built. There may be several suppliers building skid systems at their facility for delivery to the end user. There may be one main supplier that receives a majority of the components, some of which are used for building panels and skid systems, while others will be installed at the end user's facility. For another project, all components are delivered directly to the end user's facility. Depending on the logistics, some testing may be performed at the supplier's location, even by the supplier, if properly trained. Other testing will be performed at the end user's facility.

The flowchart below illustrates instrument commissioning activities covered in topics 32-1 and 32-2.

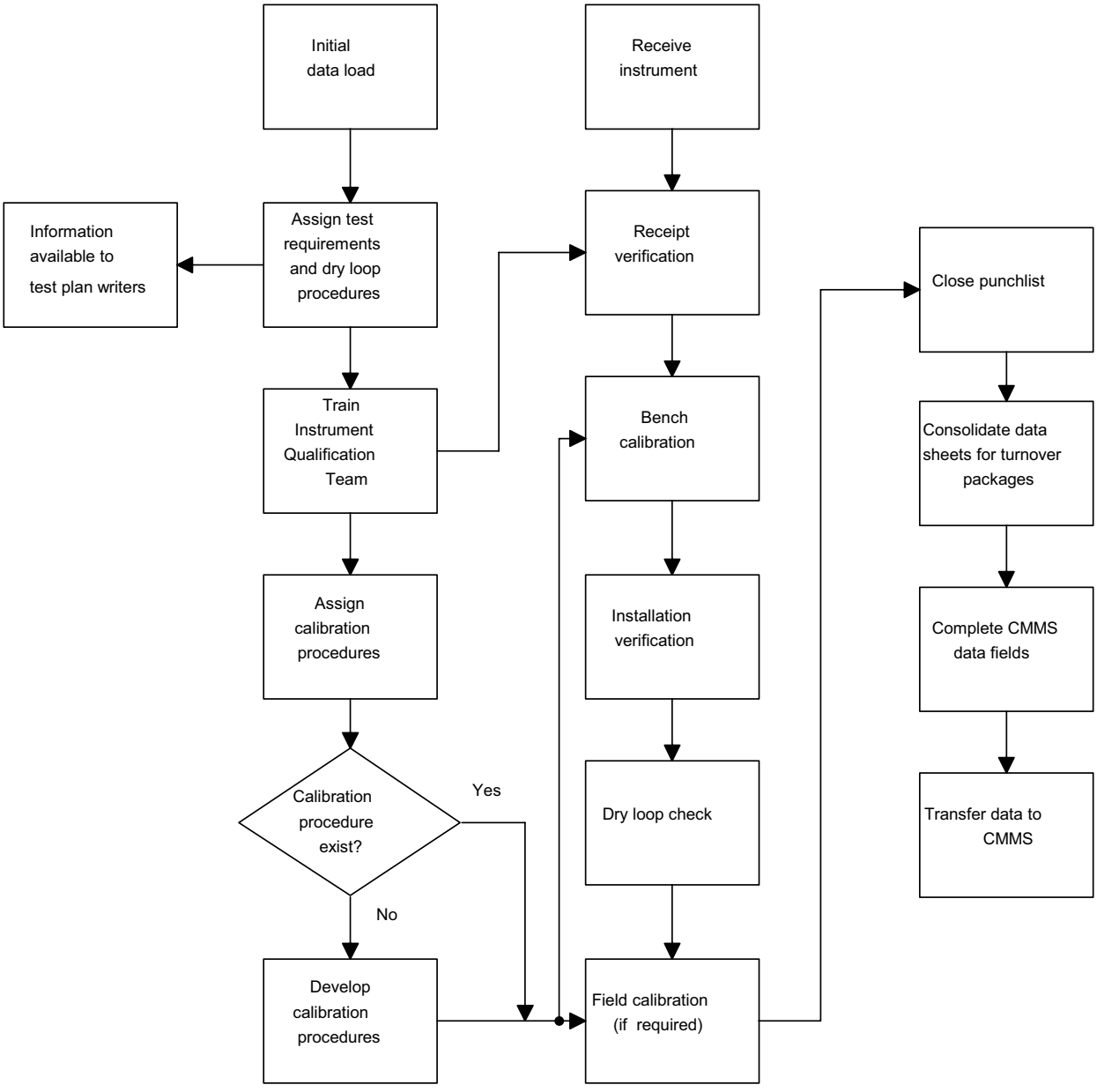


Figure 32-1: Instrument Commissioning Flowchart

32.2 Instrumentation Commissioning

To begin to build a case, we need to gather evidence that the components will work once they are installed and integrated into the system. At the component level, the first opportunity to gather this evidence occurs when the component is received. Once receipt verification is completed, a bench calibration of calibrated devices can be performed prior to installation. Once installed, instruments should be verified for proper installation. Once all wiring and system connections have been completed and the system has been powered up, loop checks and field calibrations can begin. Preferably, the control system is installed and the control programs are loaded when loop checks are performed. If, for example, the PLC code is not loaded, all testing to verify the proper indications at the human-machine interface (HMI) would need to be duplicated later.

Not all components require all testing. For example, it might make sense to perform all tests for a pressure transmitter, but none of the testing for an alarm light mounted on a panel. You might skip receipt verification and only perform installation verification for a solenoid valve because it is an off-the-shelf common item that would be simple to replace later, if defective. A testing matrix to identify the instrument commissioning activities for each element of the instrumentation and control system should be developed. All instruments and I/O should be accounted for in the testing matrix (i.e., all the diamonds and bubbles for each piping and instrumentation drawing [P&ID]). The I/O listing should be used to verify all control system inputs and outputs are accounted for. Organize the project in a way that makes sense. For example, organize projects with more than one system by P&ID. Use a database program, preferably interfaced with the overall project database, to provide an efficient means of entering the instrument information, required commissioning activities, print test forms, and generate status reports.

A simple example of an instrument commissioning testing matrix is illustrated below.

Table 32-1: Example Instrument Commissioning Test Matrix

Tag#	Description	RV (Yes/No)	IV (Yes/No)	Loop Check (Proc#)	Cal (Proc#)	Vendor, Bench, Field	Loop Tuning (Yes/No)
AIT-1002	Tank 1000 Conductivity Transmitter	Y	Y	AI	CAL-08	F	N
AIC-3453	Purified Water RO unit pH Controller	Y	Y	AI	CAL-07	F	Y
AY-3453	Purified Water RO unit pH Control I/P	Y	Y	AO	CAL-02	F	w/AIC
FT-2505	FRX 2500 Oxygen Flow Transmitter	Y	Y	AI	Vendor	V	N
HS-4800	BRZ-4800 Agitator Hand-Off-Auto Switch	N	N	DI	N/A	N/A	N
LIT-4001	WFI Tank Level Transmitter	Y	Y	AI	CAL-05	B, F	Y
LY-4001	WFI Tank Level Control I/P	Y	Y	AO	CAL-02	F	w/LIT
LCV-4001	WFI Tank Level Control Valve	N	Y	w/LY	w/LY	w/LY	w/LY
MIT-5108	Tablet Coater Dew point Indicator/Transmitter	Y	Y	AI	CAL-12	F	Y
MY-5108	Tablet Coater DP Control Damper Actuator	N	Y	AO	CAL-18	F	w/MIT
PI-1004	Tank 1000 Pressure Gauge	Y	Y	N/A	CAL-04	B, F	N
PIT-1007	Tank 1000 Return Loop Pressure Transmitter	Y	Y	AI	CAL-03	B, F	Y

PY-1007	Tank 1000 Return Loop Pressure Control I/P	Y	Y	AO	CAL-02	F	w/PIT
PCV-1007	Tank 1000 Return Loop Pressure Control Valve	N	Y	w/PY	w/PY	w/PY	w/PIT
PSE-1010	Tank 1000 Rupture Disk	N	Y	N/A (unless alarmed)	N/A	N/A	N
PSV-8111	Compressed Air Relief valve	N	Y	N/A	Vendor	V	N
SIC-4809	BRX-4800 Agitator Speed	Y	Y	AI	CAL-20	F	N
SV-3080	Clean Steam to Bioreactor Suite Control Valve Solenoid	N	Y	DO	N/A	N/A	N
TE-4003	WFI Tank Temperature RTD	Y	Y	w/TT	w/TT	w/TT	w/TRC
TT-4003	WFI Tank Temperature Transmitter	Y	Y	AI	CAL-01	B, F	w/TRC
TRC-4003	WFI Tank Temperature Recorder Controller	Y	Y	w/TT	CAL-10	B, F	Y
TY-4003	WFI Tank Temperature Control I/P	Y	Y	AO	CAL-02	F	w/TRC
TCV-4003	WFI Tank Temperature Control Valve	N	Y	w/TY	w/TY	w/TY	w/TRC
WIT-200	Weigh/Dispense Weight Transmitter	Y	Y	AI	CAL-14	F	N
XV-3080	Clean Steam to Bioreactor Suite Control Valve	N	Y	w/SV	N/A	N/A	N
ZSO-3080	Cln Stm to Brx Suite Control Vlv Open Limit Switch	N	N	DI	N/A	N/A	N
ZSC-3080	Cln Stm to Brx Suite Control Vlv Closed Limit Switch	N	N	DI	N/A	N/A	N

32.2.1 Receipt Verification (RV)

The main objectives of performing receipt verification (RV) are to verify that the device received is the device that was ordered, the device meets the specification for that instrument, and the correct vendor manuals are received for the device. For most projects, instrument specifications are developed and approved prior to purchase. Instrument specifications are developed using ISA-TR20.00.01-2001 - *Specification Forms for Process Measurement and Control Instruments, Part 1: General Considerations – Updated with 20 New Specification Forms in 2004*, or equivalent. The purchase order should also be referenced in case any additional requirements are listed there.

Receipt verification is performed upon receipt of the instrument at the end user's site, supplier's site, or off-site storage location. RV is performed per an approved RV procedure and documented on an approved data sheet printed out as a report from the inputted database information, if applicable. The following minimum requirements should be performed during RV:

- Instrument matches Purchase Order and Instrument Specification
- Manufacturer and model number verified
- Serial number and other nameplate data recorded
- Permanent tag is verified correct and properly applied
- Correct quantity of manufacturers manuals received and logged in

- Any deficiencies noted on RV Data Sheet. Any deficiencies not corrected are added to punchlist

The technical manuals should be organized for turnover to the end user. A good way to organize manuals is by manufacturer and model number. Once the RV is complete, properly disposition the device to bench calibration, designated storage location, or installation.

That being said, RV is optional. It is very convenient to perform the activities listed above at receipt. Even if RV is not performed for most devices, it should be performed for long-lead-time devices to prevent significant delays later. If RV is not performed, all the RV requirements listed above should be performed with the installation verification

32.2.2 Installation Verification (IV)

Installation verification (IV) is performed to verify the instrument or device is installed in accordance with project drawings, customer specifications, and manufacturer's instructions. Project drawings may include instrument installation details and P&IDs. For example, it is very important to verify proper orientation for some sensors, such as flow and pressure instrumentation. Pneumatic valves must have air connected to the correct port for proper operation and fail position.

To minimize duplication of effort, IV will typically be performed once all instruments in a system have been installed and authorization to begin has been received from the project manager. Once IV has started, all installation changes must be communicated to the commissioning team. If the start of IV is not communicated, undocumented changes may continue to occur during construction even after the IV has been completed. The following minimum requirements should be performed during IV:

- Instrument is installed per project drawings (i.e., installation detail, P&ID) and manufacturer's instructions.
- Instrument is properly tagged.
- Instrument wiring is properly terminated.
- Instrument air is properly connected, if applicable.
- Instrument is installed in proper location.
- Instrument can be removed for periodic maintenance and calibration (i.e., slack in flexible conduit, isolation valves, RTDs installed in thermowell) Note any discrepancies and whether the discrepancy is a deviation from specification or an observation.
- Any deficiencies should be noted on IV Data Sheet. Any deficiencies not corrected are added to punch list.

32.2.3 Loop Checks

An instrument loop is a combination of interconnected instruments that measure and/or control a process variable. An instrument loop diagram is a composite representation of instrument loop information containing all associated electrical and piping connections. Instrument loop diagrams are developed in accordance with standard ISA-5.4-1991 - *Instrument Loop Diagrams*. Minimum content requirements include:

- Identification of the loop and loop components
- Point-to-point interconnections with identifying numbers or colors of electrical and/or pneumatic wires and tubing, including junction boxes, terminals, bulkheads, ports, and grounding connections
- General location of devices such as field, panel, I/O cabinet, control room, etc.

- Energy sources of devices
- Control action or fail-safe conditions

Loop checks are performed for every I/O point. Loop checks are performed to verify each instrument loop is connected properly, indications are properly scaled, alarms are functioning properly, and fail positions are properly configured. A formalized loop check should be documented prior to placing any loop in service. This formalized program should include verifying installation against the loop diagram and simulation of signals to verify output responses and indications throughout the range.

Why is this important? A significant percentage of instrument loops have some problem, some of which would result in hidden failures. As an example, a temperature transmitter output wired to a PLC analog input provides a temperature display on an operator interface. The transmitter is calibrated from 0-100°C to provide a proportional 4-20 mA output. If the PLC programmer writes the code for this input as 0-150°C for a 4-20 mA input and a loop check is not performed, an inaccurate displayed value will result (and possibly improper control action).

Other typical problems found and corrected by performing loop checks include wiring connected to the wrong points, ground loops, and broken wires. Whenever possible, loop checks should be performed at the same time as the field calibration for devices in the loop. The same test equipment will be utilized and some of the same test connections will be used. This makes more efficient use of time and resources.

There are a number of ways to perform loop checks. To perform loop checks with maximum effectiveness, they should be coordinated with the field calibration requirements and should be performed with the control system program, such as programmable logic controller (PLC) code or distributed control system (DCS) program, completed and loaded.

Why is this important? Consider the following.

Let's say a field calibration of a temperature transmitter was performed with the RTD connected and placed in a temperature block. The 4-20 mA transmitter output was checked over the calibrated input range of 0-100°C. No remote indications or alarms were checked during the calibration. That is perfectly normal. There are a few options for the loop check. The loop check could have been performed at the same time as the calibration with the remote indications at the HMI verified and alarms checked. Or, if the field calibration was already completed, the loop check could be performed by using a milliamp simulator connected from the transmitter output to verify remote indications and alarms.

However, if no field calibration is required because the bench calibration was the only calibration requirement, you would have to start the loop check at the RTD to verify all loop components are working together. Of course, it is very important to properly reconnect the loop once the loop check is completed, or the whole thing is null and void.

Let's take the other issue where the control system program is not loaded when performing loop checks. This has been common in my experience, so it takes excellent planning to make sure the program is ready when it's time for loop checks. If the program is not ready and loop check is performed by verifying the correct PLC input displays the correct bits, we have not checked the whole loop. In too many cases I've had to do loop checks multiple times, once to the control system input (just so we could show progress) and again later from the input to verify HMI indications. A little planning would have saved time and money (and a lot of complaining).

Loop checks can be divided into four main categories: analog input, analog output, discrete input, and discrete output. Analog refers to devices that accept or deliver signals that change proportionately (includes analog signals that have been digitalized). Discrete refers to on-off signals. Examples of each signal type include:

Discrete Input (DI) - pushbuttons, switches, valve position feedback

Discrete Output (DO) - solenoids, alarms

Analog Input (AI) - process parameters such as temperature, pressure, level, flow

Analog Output (AO) - control output to an I/P or proportional valve, retransmitted analog input

Other loop checks may include RTD input and block valve. The RTD input is an analog input but may require a different test procedure and form. The block valve loop check *can be used for efficiency to test the solenoid valve and valve position inputs all together.*

RTD Input (RTD) - RTD input direct to RTD input card without the use of a transmitter

Block Valve (BV) - includes testing of solenoid, valve, and valve position feedback switch(es) in one loop check, if desired. (Otherwise, it's OK to do a discrete output check for the solenoid and valve with discrete input checks for the valve position switch(es).

For additional information on loop checks, refer to *Loop Checking: A Technicians Guide*, by Harley M. Jeffery, a part of ISA's Technician Series (2005).

32.2.4 Calibration

The ISA definition of calibration is "a test during which known values of measurand are applied to the transducer and corresponding output readings are recorded under specified conditions." To perform a calibration, a test standard is connected to the device input and the input is varied through the calibration range of the instrument while the device output is measured with an appropriate test standard. Once the as-found readings have been recorded, the device is adjusted until all readings are within the required tolerance and the procedure is repeated to obtain as-left readings.

There are a few different times during the project when calibration can be performed. First of all, the instrument vendor can calibrate the instrument to the required specification and provide calibration documentation. In this case, a certificate simply stating the device has been calibrated is not sufficient. A calibration report including calibration data, test standards used, procedure reference, technician signature and date must be included as a minimum. A bench calibration can be performed upon receipt or just prior to installation. A field calibration can be performed once installed and integrated with the instrument loop. A field calibration can be performed at the same time as a loop check, described previously, for increased efficiency. Depending on the instrument and end user requirements, a vendor calibration, bench calibration, and field calibration may be performed for some instruments with only a vendor calibration for other instruments. Here are some suggestions for when to calibrate for commissioning of a new project:

Vendor Calibration: The calibration parameters are specified prior to order and bench calibration will typically not be performed for this project.

Bench Calibration: Vendor calibration is not performed and/or it is the end-user's practice to perform bench calibrations.

Field Calibration: Always, unless it is impossible to access the instrument safely in the field or it is the end-user's practice to perform bench calibrations.

Refer to ISA's *Calibration: A Technician Guide* (2005) for examples of calibration procedures and additional information on the various elements of calibration.

32.3 Software Testing

32.3.1 Software Development

Software should be developed by the supplier in accordance with approved software development standards and a Software Quality Assurance program. These documents should detail programming standards, version control, internal testing during development, documentation of bugs and corrective actions taken to debug.

32.3.2 Program Code Testing

Prior to deployment, the functional elements of the programming code should be testing using simulations. The simulations can be performed by forcing inputs internal to the program and observing outputs. If practicable, it is better to use an external control system simulator setup that can be used to provide simulated inputs and observe outputs. Using a simulator prevents modifying the program for testing purposes, which may lead to errors if the program is not restored to original.

The testing is significantly less demanding if the programming is developed using functional blocks for each specific control actions. For example the ANSI/ISA-88.01-1995 - *Batch Control* standard uses operations or phases in procedural model and equipment logic which can be duplicated for each similar program function. This way, once the functional block is tested once, it does not need to be repeated for other program code that uses the same functional block.

32.3.3 Security Testing

Adequate controls must be placed on process control system programs and configurable devices to prevent unauthorized and undocumented changes. The controls are specified during program development or device design. Elements of system security that should be tested at a minimum are login security and access levels.

Login security ensures the appropriate controls are in place to gain authorized access to the program and to prevent unauthorized access. This can be a combination of a user ID and password or use of biometrics such a fingerprint or retina scan. Examples of login security include passwords that can be configured for a minimum number of characters, use of both alpha and numeric characters, force password change at initial login, and require password change at specified intervals. Access should be locked out for any user that cannot successfully log in after a specified number of attempts. All of the specified configurations should be challenged during testing.

Each user should be assigned an access level based on his or her job function associated with the program or device. Examples of access levels include Read-only, Operator, Supervisor, Engineer, and Administrator. Each access level would have the ability to view, add, edit, delete, print, create reports, etc., as applicable to their job function. All of the access levels would be challenged during testing.

32.4 System Level Testing

To continue building a case, we need to gather evidence that the components are working together when integrated with the equipment and control systems. At the system level, the first opportunity to gather this evidence is when the instrumentation, controls, equipment, and utilities have been connected, powered up, and turned over from construction.

32.4.1 Alarm and Interlock Testing

Alarm and interlock testing is performed to verify all alarms and interlocks are functioning properly and activate at the proper setpoints/conditions. Many of the alarm and interlock tests could be performed with loop checks, since most alarms and interlocks are activated from some loop component or control system output. If alarm and interlock tests are performed separately from loop checks, they are

documented as part of the Factory Acceptance Test, Site Acceptance Test, or operational qualification test.

32.4.2 Wet Testing and Loop Tuning

Now that we have evidence the specified components are properly installed and calibrated, the loops components are communicating, and the programming has been developed using appropriate quality procedures and tested, we can make a final case by testing the integrated operation of the system. Loop tuning would also be performed with wet testing.

Loop tuning is performed to optimize loop response to setpoint changes and process upsets. Controller **PID** parameters of **P**roportional band (or gain), **I**ntegral, and **D**erivative are adjusted to optimize the loop response and minimize overshoot. Various methods have been developed to perform loop tuning, including the Ziegler-Nichols and trial-and-error method. ISA has several resources for additional information on loop tuning, such as *Tuning of Industrial Control Systems* (Corripio) and *Maintenance of Instruments and Systems* (Goettsche).

Detailed test procedures must be used so wet testing is performed safely. The system must be in operational condition with all utilities connected, and filled with an appropriate medium, if applicable.

Each system should be tested against the specified Sequence of Operations for startup, shutdown, and operations. As an example, the system should be started up and all aspects of the startup sequence verified. In some cases simply pressing the ON button is all that is required to bring the system up to normal operation. In other cases, operator intervention is required during the startup sequence. In either case, all specified operations must be verified.

Once sequence of operation testing is performed, the system should be tested for normal operation, process setpoint changes, process upsets, and abnormal operations (to verify it recognizes improper operating parameters and places the system in a safe condition). Examples include:

Normal operation - Tank level drops to the low fill setpoint and initiates a tank filling operation

Process setpoint change - temperature setpoint is changed from 40ºC to 80ºC

Process upset or disturbance – heat exchanger steam pressure decreases

Abnormal operations – valve(s) out of position, pump not running, filter is clogged

32.5 Factory Acceptance Testing (FAT)

Factory acceptance testing (FAT) is a major milestone in a project when the system has been built, the supplier is ready to deliver, and the end-user has an opportunity to review documentation and witness performance testing. FAT is used to verify the system hardware, configuration, and software has been built, assembled, and programmed as specified.

Whenever appropriate, supplier documentation deliverables should be reviewed and testing conducted at the supplier site prior to system delivery. This will allow for troubleshooting and problem resolution prior to shipment, providing a higher level of assurance that the system will meet specifications and function properly upon delivery. In addition, problems found and corrected at the supplier site can be corrected at less cost and with less impact on project schedule. FAT should be performed with end-user representatives from various departments such as manufacturing, engineering, maintenance, information technology, and quality. During testing, systems should be challenged to best simulate actual production conditions. FAT test plan and procedures should be developed by the supplier and approved by the end user prior to testing. Some of the activities to consider during FAT are described below.

Documentation

Perform a formal review of all project deliverables such as:

- Design specifications
- P&IDs
- Instrument index
- Instrument specifications
- Instrument location drawings
- Loop diagrams
- Instrument installation details
- Sequence of events
- Logic diagrams
- DCS/PLC program
- Operating instructions
- Process flow diagrams
- Instrument, equipment, component technical manuals

Software

No matter how much review and testing of the programming is performed, it is impossible to test it all. The most efficient use of resources for is to verify:

- Compliance with Software Development Quality Assurance
- Compliance with software programming standard
- Test against software design specifications
- Security and critical functions should be tested during FAT, SAT, commissioning, and/or qualification testing
- Generate and review a sampling of historical data, trend data, and reports for adequacy

Operator Interface

The end user's ability to interact with the system depends heavily on the effort taken to ensure ergonomic and practical implementation of the operator interface. The following items should be reviewed for each operator interface:

- HMI screen layouts
- Usability
- Readability
- Responsiveness

Hardware

Hardware testing is performed to verify the system has been built according to approved hardware design specification. In addition, review completed documentation for any instrument commissioning, software testing and system level testing described in sections 32.1 and 32.2 that has been performed.

If at all possible, and many times it is, startup and operate the system. Many suppliers now have the facilities to connect temporary utilities so the system can be operated as if it were installed at the end-user's facility. Take advantage of this opportunity. Any testing completed at the FAT can significantly reduce the testing requirements at the end-user facility which can reduce the burden on an already tight schedule.

32.6 Site Acceptance Testing (SAT)

The Site Acceptance Test demonstrates the system is working in its operational environment and interfaces with instruments and equipment from other suppliers. The SAT normally constitutes a repeat of elements of the FAT in the user's environment plus those tests made possible with all process, field instruments, interfaces, and service connections established. A repeat of all FAT testing is not necessary. After all, one of the purposes of FAT is to minimize the testing required at the end-user's facility. Obviously, for systems built at the end-user facility, a FAT is not performed. All elements mentioned previously in the FAT should therefore be performed during SAT. A Site Acceptance Test Plan should be co-developed by the supplier and end-user prior to testing. Some of the activities to consider during SAT are described below:

- Repeat critical FAT elements possibly compromised by disassembly, shipment, and reassembly
- Perform testing that is now made possible with all process, field instrumentation, interfaces, communications, and service connections established
- Interface testing of critical elements of Level 3 System (e.g., MES), if applicable
- Interface testing with critical elements of Level 4 System (e.g., ERP), if applicable.

32.7 Safety Considerations

It is worth mentioning safety considerations during commissioning and system testing. Automation professionals who are not permanently assigned to a manufacturing location are typically involved with startup, commissioning, and system testing. In addition, systems are placed in unusual configurations and construction activities are usually still going on during commissioning. Also, more equipment failures occur when initially placed in service, before they are "broken in." For these reasons, more incidents occur during checkout and commissioning. Of course OSHA regulations must be followed, but let's mention a few common sense safety precautions for those engineers that don't often get out in the field.

Electrical safety – Technically, any voltage source over 30 volts can kill you. During construction and startup, electrical panels are routinely left open. You should always assume the wires are live until proven otherwise. Taken adequate precautions when working in and around control panels that are energized. For example, remove all metal objects (watches, jewelry), insulate surrounding areas, wear rubber sole shoes, and never work alone. Always have someone working with you (and not in the panel with you!). Depending on the voltages present and risk involved it may be a good idea to tie a rope around yourself so the other person can pull you out, just in case.

Pressurized systems – Precautions must be taken for any system or component that could be under pressure. During startup, systems are turned over from construction in an unknown status. Valve line-up checks should be performed to place the system in a known condition. Even then systems are placed

in unusual operating conditions to perform testing. Always know what you're working with and always proceed with caution when manipulating system components and changing system conditions. As mentioned before, if a component is going to fail, it will tend to fail when initially placed in service. I've seen, on more than one occasion, tank rupture disks fail the first time a system is brought up under steam pressure. This is even after the rupture disk was visually inspected for defects during installation verification. When those things blow, you better hope you are not in its discharge path. Check system pressure using any available indications before removing a component that would breech system integrity. However, even if a gauge reads 0 psig, carefully remove the component. The gauge could be faulty. And again, **never work alone**. Make sure somebody knows where you are working.

Extreme temperature – High- and low-temperature systems, such as steam, hot water, cryogenic storage, and ultra low temperature freezer systems can be very dangerous even if everything is kept inside the pipes, tanks, and chambers. Although insulation minimizes burn risks, there is exposed piping. Do not grab any exposed piping until you know it is safe to touch. We learned this as a kid the first time we touched a hot stove. We seem to have to relearn this for every new project. Even the smallest steam burn hurts for several days. Exposure to cryogenic temperatures such as liquid nitrogen is just like frostbite and steam burn and hurts just as badly.

There are also dangers with using liquid nitrogen in small, enclosed spaces. The nitrogen will displace the air and cause you to pass out and eventually die. The bottom line is to **know what you're working with, be smart, and never work alone**.

32.8 References

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About the Author

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